

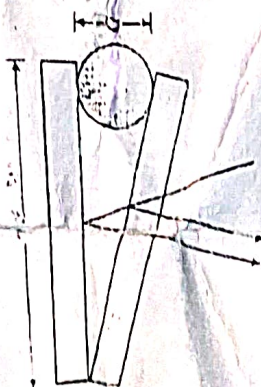
TIME: 40 Min

MAX. MARKS :10

Marks BL

- 1 Define Malus law. An unpolarized light beam of intensity I_0 passes through three successive polaroids P_1 , P_2 and P_3 . The polarization axis of P_2 makes an angle 60° with P_1 and that of P_3 makes an angle of 30° with P_2 . What are the corresponding intensities of light (I_1 , I_2 and I_3) coming out from P_1 , P_2 and P_3 ?
- 2 Briefly explain the phenomenon of Birefringence (double refraction). Linearly polarized light of wavelength 525 nm falls, at normal incidence, on a birefringent crystal that is cut with its faces parallel to the optic axis. What is the smallest possible thickness of the crystal if the emergent ordinary ray (o-ray) and extra-ordinary ray (e-ray) combine to form linearly polarized light? Given the refractive indices for the o-ray being ($n_o = 2.356$) and that of e-ray is ($n_e = 2.378$).

- 3 Write down the conditions for constructive and destructive interference in a thin wedge shaped film. Identify all the terms. A broad-source of light ($\lambda = 680 \text{ nm}$) illuminates normally 2 glass plates of length, $L = 120 \text{ mm}$ long (in air) that touch at one end and are separated by a wire of diameter, $D = 0.048 \text{ mm}$ at the other end (See Figure). Calculate:
- (a) Fringe width for bright fringe
- (b) Number of bright fringes over the entire length of the plates



BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION)

CLASS: BTECH / IMSC
BRANCH: BT/CE/CHEMICAL/ME/PIE/FT/PHYSICS

SEMESTER: I
SESSION: MO/2025

SUBJECT: PH24101 PHYSICS

TIME: 02 HOURS

FULL MARKS: 25

INSTRUCTIONS:

1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., If applicable, will be supplied to the candidates

- | | | | CO | BL |
|--------|--|-----|----|-----|
| Q.1(a) | Define Brewster's law. Light travelling in water ($n_1 = 1.33$) is incident on a glass plate of ($n_2 = 1.53$). At what angle of incidence (in degrees), is the reflected light completely linearly polarized? | [2] | 1 | 1,3 |
| Q.1(b) | Prove that at angle of incidence equal to Brewster's angle, the reflected and refracted rays are perpendicular to each other. | [3] | 1 | 2 |
| Q.2(a) | Explain the phenomenon of interference in thin film of uniform thickness and obtain the conditions of constructive and destructive interference. | [2] | 1 | 1,2 |
| Q.2(b) | A plane wave ($\lambda = 500 \text{ nm}$) falls normally on a long narrow slit of width 0.5 mm . Calculate the angles of diffraction (in degrees) corresponding to the first and second minimum. How will the angles change if the slit width is reduced to 0.1 mm ? | [3] | 1 | 3 |
| Q.3(a) | Define Gauss law of electrostatics. If the di-electric displacement vector (in vacuum) is given by $\vec{D} = (2y^2 + z)\hat{i} + (4xy)\hat{j} + (x)\hat{k}$, find the volume charge density at the point $(-1, 0, 3)$ in Coulomb/m ³ . | [2] | 2 | 1,3 |
| Q.3(b) | Use Gauss law to find the electric field vector ($\vec{E}$) inside and outside a uniformly charged sphere of radius R . Given the volume charge density of the sphere is ρ_0 Coulomb/m ³ . | [3] | 2 | 3 |
| Q.4(a) | Define dielectric polarization ($\vec{P}$). How is it related to the electric field ($\vec{E}$) for a linear, homogeneous, isotropic di-electric? | [2] | 2 | 1,2 |
| Q.4(b) | A sphere of radius R carries a polarization $\vec{P} = k\vec{r}$ where k is a constant and $\vec{r}$ is the radial vector from its center. Calculate:
(i) The bound surface (σ_b) and volume (ρ_b) charge densities.
(ii) The Electric field ($\vec{E}$) inside and outside the sphere | [3] | 2 | 2,3 |
| Q.5(a) | Define an inertial frame of reference. What are the postulates of Special theory of relativity | [2] | 3 | 1 |
| Q.5(b) | Derive the Lorentz transformations connecting two inertial frames moving with a velocity v with respect to each other | [3] | 3 | 3 |

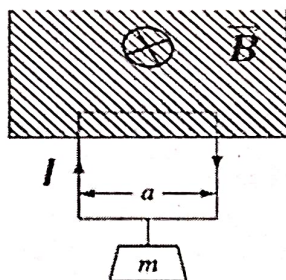
:::::16/10/2025:::::M

INSTRUCTIONS:

1. The question paper contains 5 questions each of 10 marks and total 50 marks.
 2. Attempt all questions.
 3. The missing data, if any, may be assumed suitably.
 4. Before attempting the question paper, be sure that you have got the correct question paper.
 5. Tables/Data handbook/Graph paper etc. to be supplied to the candidates in the examination hall.
- Useful Info.: Electron mass = 9.11×10^{-31} Kg, Speed of light (c) = 3×10^8 m/s, $1 \text{ eV} = 1.6 \times 10^{-19}$ J, Planck's constant (h) = 6.63×10^{-34} Js, $\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ Fm}^{-1}$

- Q.1(a) Draw the schematic of Newton's rings setup. Use it to derive the expressions for the radius of the m^{th} order dark and bright rings respectively. [5] CO 1,2
- Q.1(b) What is the difference between interference and diffraction? [5] CO 1,3
A uniform thin film of thickness $t = 4 \times 10^{-5}$ cm and refractive index $\mu = 1.5$ is illuminated normally by white light. Of what color will the thin film appear in reflected light?

- Q.2(a) Define Ampere's law. [5] CO 2, 1,3
A rectangular loop of wire supporting a mass m , hangs vertically with one end in a uniform magnetic field B , which points into the page (See Fig. below). For what current I in the loop, would the upward magnetic force exactly balance the gravitational force downward?



- Q.2(b) Write down Maxwell's equations in free space and name each one of them. [5] CO 2, 1,3
Apply Gauss law to calculate the electric field vector ($\vec{E}$) at point P which is at a distance Z above an infinite line charge of linear charge density λ .
- Q.3(a) Define relativistic momentum. Write down the formula for relativistic kinetic energy. [5] CO 3, 1,1,2
Derive the relativistic energy-momentum relationship.
- Q.3(b) Using special theory of relativity, show that a particle of zero rest mass must always travel at the speed of light. [5] CO 3, 2,3

A cosmic ray particle has a lifetime of 10^{-7} s when it is at rest, how far does it travel (in the earth frame) before decaying, if its speed is $0.99c$ when it is created?

- Q.4(a) Write down the Schrodinger equation for a particle in a 1-dimensional infinite potential box of length L : [5] CO 4, 1,2,3

$$V(x) = 0, \text{ if } 0 < x < L$$

$$V(x) = \infty, \text{ if } x \leq 0 \text{ or } x \geq L$$

Solve it to obtain the normalized wave function and energy eigenvalue for the n^{th} energy level.

Use these expressions to calculate the ground state energy of an electron (in eV) confined to a 1-Dimensional infinite potential box 0.1 nm wide.

PTO